MDS482-4 PROJECT

PAWMAP

A Community-Driven Stray Animal Rescue and Welfare Platform

Submitted By

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Scope

PawMap is a mobile and web-based application designed to coordinate the rescue, medical care, fostering, and adoption of stray and abandoned animals across urban and semi-urban communities. The platform integrates real-time geo-mapping, volunteer management, veterinary coordination, and community-driven reporting into a unified ecosystem for animal welfare. It serves a diverse set of stakeholders including the general public, community volunteers, registered non-governmental organisations, veterinary clinics, and prospective adopters, with access levels and features tailored to each role.

Need

India has an estimated 62 million stray dogs alongside a substantial population of stray cats, with abandonment remaining a persistent and growing concern in urban areas. Existing rescue efforts are largely fragmented, with poor coordination between volunteers, NGOs, and veterinary professionals. Reported cases frequently go unattended due to the absence of visibility, accountability, and structured follow-up mechanisms. No unified platform currently exists that combines real-time case tracking, volunteer dispatch, animal medical records, and adoption pipelines in a single interface. PawMap addresses this critical gap with a structured, technology-driven solution that reduces response times, eliminates duplication of effort, and ensures continuity of care for every animal from the moment it is spotted to the point of resolution.

Objectives and Expected Outcomes

The primary objectives of PawMap are to enable any community member to report a stray or injured animal in under 30 seconds using location tagging and photographs; to coordinate volunteer response through geofenced notifications and a conflict-free, atomic case-claiming system; to maintain a longitudinal medical record for each animal across all caregivers and veterinary interactions; and to facilitate structured foster-to-adopt pipelines that lower barriers to permanent placement. Expected outcomes include a measurable reduction in unattended stray cases within participating localities, increased volunteer engagement through transparent reliability scoring, improved continuity of medical care, and a higher rate of successful adoptions facilitated through verified adopter queues and shareable animal profiles.

Key Features

PawMap is organised into five functional modules. The Core Map and Reporting module enables users to pin stray animals with a photograph, GPS coordinates, and a five-point temperament tag, tracking each case through a live status pipeline from Spotted to Resolved; an SOS Mode allows one-tap urgent alerts to all nearby volunteers and NGOs. The Volunteer Coordination module manages geofenced case notifications, atomic claiming to prevent duplicate response, a waitlist queue with automatic reassignment after 30 minutes, and a volunteer reliability index derived from response rate and case completion history. The Veterinary and Medical module provides a mapped directory of clinics, a per-animal medical history log recording vaccination, sterilisation, and deworming status, and a vet case-claiming system for serious injuries. The Adoption and Rehoming module offers shareable animal profile cards, a structured adopter interest queue, a foster-to-adopt conversion pipeline, and a community adoption success feed.

Beneficiaries

The primary beneficiaries of PawMap are stray and abandoned animals who receive timely rescue, appropriate medical attention, and pathways to permanent adoption. Secondary beneficiaries include community volunteers who gain structured tools for coordinated and impactful engagement, NGOs and animal welfare organisations that benefit from automation, transparency, and reporting capabilities, veterinary professionals who can update and retrieve animal health records in real time, and members of the general public who wish to report animals in need or offer adoption. Local municipal bodies stand to benefit indirectly from spatial abandonment data that can inform urban animal welfare policy and targeted intervention planning.

Technology Stack

PawMap will be developed as a web application using Django and Django REST Framework for the backend, providing core application logic and a RESTful API layer that connects the React (with Bootstrap 5) frontend to the database and machine learning components. Data will be stored in PostgreSQL, extended with PostGIS for spatial data storage and querying, enabling efficient geospatial operations such as proximity search and clustering. Background task scheduling — including geofenced alerts, auto-release of unclaimed cases after 30 minutes, and SOS escalation — will be handled using Celery with Redis as the message broker. Real-time geolocation and mapping functionality will be implemented via the Google Maps API, while push notifications for volunteer dispatch and SOS alerts will be delivered through Firebase Cloud Messaging.